

## EDUCATION

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- **Bachelor of Engineering in Automation & Robotics** Expected July 2026  
*Modern Education Society College of Engineering* Pune, Maharashtra
  - **Relevant Coursework:** Robotics IIT-K (NPTEL) & CS50P (Python)

## EXPERIENCE

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- **Robotics Intern (Industrial Robotics)** Onsite  
*FANUC India Pvt. Ltd.* Feb 2026 – Present
  - Worked with FANUC industrial and collaborative robots including CRX-10iA/L, M-2000iA, M-900iB, and R-2000iC series on simulation, inspection, palletizing, and machine tending applications.
  - Performed robotic feasibility studies and process simulations using FANUC RoboGuide for industrial automation customers involving palletizing, chip-cleaning, and multi-machine handling systems.
  - Optimized robot trajectory planning using Target Point methods, improving simulation efficiency and reducing excessive TP programming overhead.
  - Conducted thermal and joint-load analysis for high-cycle robotic operations, evaluating reducer lifespan and operational constraints under elevated ambient temperatures.
  - Worked on palletizing simulations, cycle-time optimization, EOAT orientation adjustments, and robot reachability validation.
  - Integrated FANUC RoboGuide with ROS 2 Humble environment to explore digital twin workflows and bidirectional robot-simulation communication.
  - Documented RoboGuide and ROS2 integration workflows for replication across systems using WSL2 and Ubuntu-based development environments.
  - Assisted in loan inspection, rectification, backup restoration, and commissioning workflows for CRX and LR Mate robot systems.
  - Created technical documentation and learning resources for RoboGuide onboarding and industrial robotics training.
  - Participated in customer demonstrations, factory visits, and internal process improvement activities including 5S implementation documentation.
- **UAV Engineer** Onsite  
*QuantumTechFPV* Oct - Dec, 2026
  - Core member of the engineering and build team for end-to-end development and delivery of 10 hexacopter anti-jamming UAVs with fiber-optic primary control and RF (ELRS) failover (5 km range), delivered to an Indian Army battalion in Prayagraj.
  - Performed early-stage assembly including soldering high-current bullet connectors, ESC phase wires, and 6S power distribution, ensuring low-resistance and reliable joints.
  - Developed comprehensive technical and operator manuals for a hexacopter UAV platform with fiber-optic primary control and RF (ELRS) failover, documenting system architecture, ArduPilot baseline configuration, and integration workflows.
  - Standardized pre-flight, failover, and acceptance testing procedures to improve link reliability, recovery safety, and field readiness.
  - Designed battery lifecycle management and BMS tracking framework with audit-grade logging to reduce in-field power failures.
- **Automation Intern** Onsite  
*Spark Minda* July - September, 2025
  - Gained comprehensive exposure to Surface Mount Technology (SMT) production line, learning the complete PCB assembly process from laser marking and solder paste printing through automated optical inspection and quality control.
- **Copywriter** Remote  
*cryptoprism.io* Jan - March, 2024
  - Worked as a technical content copywriter for a DEX screener SaaS product targeting cryptocurrency users, improving user engagement and product adoption.

## PROJECTS

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- **DeX: Manipulator** | *CAD, 3D Printing, ESP32*
  - Built a 5-DOF robotic manipulator with teleoperation using gesture control. Implemented wireless communication via ESP32 and flex sensors for hand tracking, exploring intuitive human-robot interaction.
- **Robot\_Mechanics** | *Python, NumPy, Matplotlib*
  - Developed an open-source Python library implementing core robotics concepts (transformations, kinematics, robot link modeling) from first principles. Designed modular components with unit tests and visualization for validation and clarity.
- **Preview (PDF Viewer)** | *PyQt5, MuPDF*
  - Built a minimalistic, cross-platform PDF viewer inspired by Evince and Mac Preview using PyQt5 and MuPDF for fast rendering.

## TECHNICAL SKILLS

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**Languages:** Python, MATLAB

**CAD Engineering Tools:** SolidWorks, LaTeX, Figma

**Systems Platforms:** Ubuntu, WSL2, Windows

**Concepts:** Digital Twins, Robot Kinematics, Trajectory Planning, Joint Load Analysis, Industrial Automation

## EXTRACURRICULAR COURSES

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- From Big Bang to Black Holes - Tokyo University (Coursera)
- CS50 Introduction to Python - Harvard University
- CS50 Introduction to Artificial Intelligence - Harvard University

## LEADERSHIP & SOCIAL ENGAGEMENTS

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- **Co-founder, Avinya Club:** Founded project-focused club for hackathons and technical development at MESWCOE Pune
- **Technical Head, ASME Student Chapter:** Led technical initiatives and projects as head of American Society of Mechanical Engineering Student Chapter at MESWCOE Pune
- **Event Organizer & Speaker:** Organized various technical events and served as keynote speaker, winning multiple competitions

## INTERESTS & HOBBIES

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**Reading:** Non-fiction, Evolution, Physics, Philosophy, Astronomy, Science Fiction

**Music:** Instrumental, Original Soundtracks

**Writing:** Technical blogs, Poetry, Personal journals